PRACTICAL EXAMINATION

Software Requirements Engineering (SRE)

Student Name: _____	Roll Number: _____
Date: _____	Section: _____

Duration: 1 Hour 30 Minutes	**Total Marks: 30**	**Attempt: All Questions**

General Instructions

- Attempt all questions.
- Complete the tasks on the computer using appropriate tools such as draw.io and LibreOffice Writer.
- Save diagrams and written responses in a single folder named with your roll number.
- Use clear labels and appropriate notation in all diagrams.
- Submit all required files before the examination ends.

Question 1 - Requirements Engineering Failure Analysis (5 Marks)

A university developed an online admission system. After deployment, students complained that they could not track their application status. The administration also discovered that the system did not support document correction, although this feature was essential to the admission department.

a. Identify two requirements engineering mistakes that may have caused these problems. (2)

b. Identify two stakeholders who should have been consulted during requirements elicitation. (1)

c. Suggest two requirements engineering activities that could have prevented these failures. (2)

Question 2 - System and Context Diagram (5 Marks)

A university wants to develop an Online Course Registration System with the following functionality:

• Students can search available courses.

• Students can register for courses.

• Teachers can view their registered students.

• The administration can manage courses.

• The system communicates with the university database.

Task: Using draw.io, construct a system context diagram. Your diagram must include:

- The system boundary.
- External entities.
- Data exchanged between entities and the system.
- Appropriate labels for data flows.

Question 3 - Requirement Sources and Kano Model (5 Marks)

A university is planning to improve its Student Portal. The following features have been proposed:

1. Secure student login.

2. Displaying examination results.

3. Fast dashboard loading.

4. Personalized course recommendations.

5. Dark mode.

a. Identify three possible sources from which requirements for this portal can be collected.

b. Classify the five features using the Kano Model**.** Briefly justify one of your classifications.

Question 4 – Requirements Elicitation Techniques (2.5 Marks)

A hospital wants to develop an Online Appointment Management System. Doctors complain about scheduling conflicts, while patients complain about long waiting times.

a. Select two appropriate requirements elicitation techniques for investigating these problems.

b. Explain why each selected technique is suitable.

c. Write one relevant question you would ask a doctor during an interview.

Question 5 - Natural Language Requirements and Requirements Template (2.5 Marks)

Consider the following requirements for an online banking application:

1. The system should be fast.

2. The system should be user-friendly.

3. The system should allow customers to transfer money.

4. The system should be highly secure.

a. Identify two problems associated with the given requirement statements. (2)

b. Rewrite any two requirements to make them more specific and testable. (2)

c. Write one requirement using a structured requirements template containing a requirement ID, description, priority, and acceptance criteria.

Question 6 – Model-Based Requirements Documentation (10 Marks)

A student wants to register for a course through a university portal. The process involves:

1. Student logs into the portal.

2. Student searches for available courses.

3. System checks course availability.

4. If seats are available, the student registers.

5. If no seats are available, the system displays an error message.

6. System confirms successful registration.

Perform the following tasks using draw.io:

a. Construct an activity diagram representing the registration process.

b. Identify the primary actor and one possible secondary actor for the corresponding use case. (1)

Submission Requirements

Create one folder named Roll_Number_SRE_Exam and save the following files inside it:

- Q1_Failure_Analysis.odt

- Q2_Context_Diagram.png
- Q3_Kano_Model.odt
- Q4_Elicitation.odt
- Q5_Requirements.odt
- Q6_Activity_Diagram.png